Labor Market Concentration, Minimum Wages, and Local Property Crime Rates*

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Motivated by a theoretical framework in which legal wages represent the opportunity cost of crime, this paper investigates how labor market concentration moderates the effect of the minimum wage on local property crime rates. I integrate a model of criminal participation into an oligopsony model of the labor market, showing that when firms have wage-setting power, a wage floor increases both employment and wages, unambiguously reducing crime. Using FBI and Lightcast data from 2010–2016, I find that in highly concentrated markets, a 1% minimum wage increase raises employment by 0.16% and reduces larceny-theft by 0.31%.

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1. Introduction

Crime imposes a significant burden on the United States economy, with estimated annual costs ranging from $4.71 to $5.76 trillion (Anderson 2021). Policymakers often debate whether broad labor market interventions, such as the minimum wage, can be used to improve economic opportunity and deter illicit activity.[1]

However, if labor markets are competitive, a minimum wage has a theoretically ambiguous effect on crime. On one hand, it increases the opportunity cost of crime for those who remain employed—the "wage effect." On the other hand, there is the risk that firms will lay off employees, which could push marginal workers into criminal behavior—the "employment effect." Because these forces operate in opposite directions, the net impact of the minimum wage on property crime depends entirely on which effect dominates. Consistent with this theoretical ambiguity, some studies have found that the minimum wage decreases crime (Agan and Starr 2018; Fernandez, Holman, and Pepper 2014), while others have found the opposite (Beauchamp and Chan 2014; Braun 2019; Fone, Sabia, and Cesur 2023; Hashimoto 1987).[2]

This theoretical ambiguity, however, hinges on the assumption of competitive labor markets. When employers have wage-setting power, a minimum wage can cause firms to *increase* employment (Azar, Huet-Vaughn, Marinescu, Taska, and von Wachter 2024). This fundamentally alters the predicted impact on crime. Because the wage and employment effects now operate in the same direction, the minimum wage should unambiguously reduce crime when labor markets are imperfectly competitive. Accounting for this market structure is crucial, as a large and growing body of evidence suggests that United States (US) labor markets are characterized by pervasive employer wage-setting power.[3]

In this paper, I develop a theoretical framework that formalizes this intuition. I integrate an economic model of crime with an oligopsonistic model of the labor market. The model yields a sharp, intuitive prediction: when labor markets are highly concentrated, the introduction of a

[1] For example, the Council of Economic Advisors (CEA) released a report recommending the use of the minimum wage to reduce property crime in 2016. See https://obamawhitehouse.archives.gov/sites/default/files/page/files/20160423_cea_incarceration_criminal_justice.pdf.

[2] This conflict parallels the long-standing debate regarding whether the minimum wage causes disemployment. For a summary, see Neumark, Salas, and Wascher (2014) or Allegretto, Dube, Reich, and Zipperer (2017).

[3] See Manning (2021) for a review.

minimum wage lacks a disemployment effect. Because employment increases, the wage and employment effects operate in the same direction, unambiguously reducing property crime.

To test this prediction empirically, I construct a novel county-level panel dataset spanning 2010 to 2016. Data on local property crime come from the FBI's Uniform Crime Reporting (UCR) program (Kaplan 2019). I focus primarily on larceny-theft, as it is the property crime most likely to respond to shifts in low-wage labor market conditions, though I also evaluate burglary and motor vehicle theft. To measure local employer concentration, I utilize a comprehensive dataset of near-universe online job vacancies from Lightcast, supplemented by the Quarterly Census of Employment and Wages (QCEW) and Local Area Unemployment Statistics (LAUS). Following the empirical strategy of Azar et al. (2024), I calculate local Herfindahl-Hirschman Indices (HHI) for three key minimum-wage occupations that also tend to recruit workers using online vacancies: cashiers, retail salespersons, and stock clerks.[4] I then employ a two-way fixed effects design to estimate how this concentration moderates the minimum wage's impact on local crime rates. Crucially, my preferred specification includes a rich set of criminal justice controls—such as clearance rates, violent crime rates, and the number of reporting police agencies—to isolate the labor market effect from unobserved shifts in local law enforcement practices or broader crime reporting trends (Braun 2019).

My empirical findings strongly support the theoretical prediction. In the most concentrated labor markets (defined as those with an HHI above 0.25), I find that a 1% increase in the minimum wage is associated with a 0.16% increase in employment and a 0.31% decrease in larceny-theft. To put these magnitudes into perspective, if the crime-reducing mechanism is driven entirely by the employment effect, the average county would experience a decrease of 1.05 larceny-theft incidents for every additional individual employed. I do not find strong evidence that the minimum wage affects burglary or motor vehicle theft. To further validate these findings, I conduct a series of placebo tests using violent crime as the dependent variable. Because violent crimes are theoretically less responsive to legal labor market incentives, they serve as a useful check against spurious correlations. The point estimates from these placebo regressions are

[4]Another key industry frequently studied in the minimum wage literature is the restaurant industry (Accommodation and Food Services). However, it is less common for restaurants to recruit workers by posting job vacancies online, as noted by Azar et al. (2024), and this makes the Lightcast vacancy data less reliable for measuring labor market concentration among restaurants. Therefore, I focus on minimum-wage retail occupations.

statistically insignificant and often carry the opposite sign, providing reassuring evidence that my main property crime results are not driven by unobservable local trends in crime reporting or policing intensity.

This paper contributes to three distinct strands of literature. First, it bridges the deeply divided literature on the minimum wage and crime cited above. The estimates in this literature vary wildly, from a 0.2% increase in youth arrests for a 1% wage increase (Fone et al. 2023) to a 2.8% decrease in prison recidivism for a $0.50 wage increase (Agan and Starr 2018). By introducing imperfect competition as a moderating variable, my paper provides an explanation for these historically conflicting estimates. To the best of my knowledge, this is the first study to demonstrate that the crime-elasticity of the minimum wage depends on local labor market concentration.[5]

Second, this paper builds upon the foundational research connecting crime, wages, and employment (Becker 1968; Ehrlich 1973). While a large literature tests the reduced-form relationships between legal labor market opportunities and crime (Gould, Weinberg, and Mustard 2002; Grogger 1998; Raphael and Winter-Ebmer 2001; Yang 2017), which Chalfin and McCrary (2017) reviews extensively, recent advances using administrative microdata have isolated the causal impact of job displacement on criminal behavior (Bennett and Ouazad 2020; Rose 2018). For example, Bennett and Ouazad (2020) find that job displacement in a mass layoff increases the likelihood of property offenses by 0.38 percentage points, and Rose (2018) finds a 30% increase in the likelihood of offending following job loss. My paper contributes to the Becker-Ehrlich tradition by quantifying the combined employment and wage opportunity-cost enhancements of the minimum wage in concentrated markets. While the simultaneous increase in wages and employment makes it difficult to cleanly disentangle the two effects, doing so remains an important avenue for future research.

Finally, my findings fit within the large literature documenting pervasive monopsony power in the US economy (Arnold, Dobbie, and Yang 2018; Azar, Marinescu, and Steinbaum 2019;

[5]Many studies in the literature find conflicting evidence regarding the effect of the minimum wage on employment (Card and Krueger 1994; Cengiz, Dube, Lindner, and Zipperer 2019; Clemens and Wither 2019; Derenoncourt and Montialoux 2021; Dube, Lester, and Reich 2010; Harasztosi and Lindner 2019; Jardim, Long, Plotnick, van Inwegen, Vigdor, and Wething 2022; Meer and West 2016; Neumark and Wascher 1992). Azar et al. (2024) show that labor market concentration is a powerful moderator that explains the diversity of effect sizes documented in the minimum wage and employment literature.

Kroft, Luo, Mogstad, and Setzler 2025; Lamadon, Mogstad, and Setzler 2022; Qiu and Sojourner 2023; Schubert, Stansbury, and Taska 2020); for a review, see Manning (2021). With recent work showing that over 60% of US labor markets are highly concentrated (Azar, Marinescu, Steinbaum, and Taska 2020), understanding the broader implications of this market structure is critical. A primary takeaway of this paper is that researchers and policymakers should not ignore imperfect competition in the labor market when studying spillover effects of labor market policies on social outcomes.

The remainder of this paper is organized as follows. Section 2 explains the theoretical framework. Section 3 outlines the data. Section 4 describes the empirical strategy. Section 5 presents the results, and Section 6 concludes.

2. Theoretical Framework

In this section, I outline a model connecting local property crime rates, minimum wages, and local labor market concentration. The framework integrates a Becker-Ehrlich model of crime (Becker 1968; Ehrlich 1973) with an oligopsony model of the low-wage legal labor market (Azar et al. 2024; Boal and Ransom 1997). All formal mathematical proofs for the propositions presented below can be found in Appendix A.

2.1. Workers' Decision Problem and Sector Selection

Consider a single local labor market populated by a unit mass of individuals. Each individual i decides between three mutually exclusive sectors: legal employment, property crime, or non-participation. Let w be the prevailing legal wage. Individual i faces a potential return to property crime, r_i, and a value of non-participation, b_i, drawn from a joint cumulative distribution function $G(r, b)$ with a continuous joint probability density function $g(r, b)$.

Individuals choose the sector that maximizes their payoff: $\max\{w, r_i, b_i\}$. The unconstrained supply of labor to the legal market, $L(w)$, is the mass of individuals for whom the legal wage dominates both alternatives:

$$L(w) = \mathbb{P}(w \geq r_i \text{ and } w \geq b_i) = G(w, w). \tag{1}$$

The unconstrained supply of property crime, $C(w)$, is the mass for whom crime dominates both the legal wage and non-participation:

$$C(w) = \mathbb{P}(r_i > w \text{ and } r_i \geq b_i) = \int_w^{\infty} \left(\int_{-\infty}^{r} g(r, b)\, db \right) dr. \tag{2}$$

By Leibniz's rule, $L'(w) > 0$ and $C'(w) < 0$, reflecting the opportunity cost effect of legal wages on criminal participation.[6]

2.2. *Firms and Labor Demand*

The demand side consists of N identical firms. Let $f(l)$ be a strictly concave production function ($f' > 0$, $f'' < 0$). Total market employment is $L = \sum_{j=1}^{N} l_j$. Firm j chooses labor quantity l_j to maximize profit, but the specification of its marginal cost of labor depends entirely on its market power. We analyze two distinct regimes: a perfectly competitive benchmark where firms are wage-takers, and an oligopsony regime where firms internalize the inverse labor supply function $W(L)$ derived from Equation (1).

2.3. *Equilibrium and the Minimum Wage*

To understand how a wage floor influences the aggregate crime rate, we first consider the standard competitive benchmark. In this setting, the policy forces a fundamental trade-off between raising the opportunity cost of crime for the employed and displacing other workers into the criminal sector.

Proposition 1 (Wage-Taking Regime)**.** *Assume firms lack market power and act as wage-takers, perceiving labor supply as perfectly elastic. The marginal effect on total crime C of introducing a minimum wage $\bar{w}$, evaluated at the market-clearing equilibrium wage w^*, is:*

$$\left.\frac{\partial C}{\partial \bar{w}}\right|_{\bar{w}=w^*} = \underbrace{C'(w^*)}_{\textit{Wage effect } (-)} + \underbrace{\Omega(w^*) \cdot [L'(w^*) - D'(w^*)]}_{\textit{Employment effect } (+)}$$

[6] Leibniz's rule for differentiation under the integral sign implies that $\frac{\partial C}{\partial w} = -\int_{-\infty}^{w} g(w, b)\, db$. Since $g(r, b)$ is a non-negative probability density, $C'(w) < 0$, reflecting the increased opportunity cost of crime.

where D is aggregate labor demand and $\Omega(w^*) = \mathbb{P}(r_i > b_i \mid w^* \geq r_i \text{ and } w^* \geq b_i)$ is the conditional probability that a rationed worker's second-best option is crime. The numerator of this probability is given by $\int_{-\infty}^{w^*} \int_{b}^{w^*} g(r, b)\, dr\, db$.
(See Appendix A for proof).

The theoretical ambiguity in Proposition 1 stems entirely from the assumption that firms operate on their marginal revenue product curve. However, low-wage labor markets are characterized by a significant degree of monopsony power (Azar et al. 2024). When firms face an upward-sloping labor supply curve, the introduction of a minimum wage fundamentally alters the marginal cost of hiring, effectively removing the incentive to under-hire. This has spillover effects on local property crime rates.

Proposition 2 (Oligopsony Regime)**.** *Assume firms engage in Cournot competition for workers, recognizing that their hiring decisions influence the market wage via $W(L)$. A minimum wage $\bar{w}$ binding within the markdown wedge ($w < \bar{w} \leq f'(L/N)$) does not induce rationing. The effect on crime is unambiguously negative:*

$$\frac{\partial C}{\partial \bar{w}} = C'(\bar{w}) < 0$$

(See Appendix A for proof).

2.4. The Role of Market Concentration

Proposition 2 establishes that a minimum wage can strictly reduce crime provided it does not exceed the marginal revenue product of labor. The likelihood of this condition holding depends on the initial size of the wage markdown wedge, which is itself a structural feature of the local market's concentration.

Proposition 3 (Concentration and Crime Elasticity)**.** *In the oligopsony regime, the magnitude of the crime-reducing effect of a minimum wage increases with labor market concentration (N^{-1}):*

$$\frac{\partial^2 C}{\partial \bar{w} \partial (N^{-1})} < 0$$

(See Appendix A for proof).

2.5. Testable Hypothesis

These theoretical insights directly motivate the empirical strategy. If employer market power mitigates the displacement effects of wage floors, we should observe heterogeneous crime responses across local labor markets depending on their competitive structure. Specifically, the theoretical framework yields the following testable prediction:

Hypothesis 1. *The crime elasticity of the minimum wage is a decreasing function of local labor market concentration, becoming unambiguously negative in highly concentrated markets (follows from Propositions 2 and 3).*

3. Data

3.1. Crime Data

Crime data come from the FBI's Uniform Crime Reporting (UCR) Program, compiled by Kaplan (2019) and accessed via the Inter-university Consortium for Political and Social Research (ICPSR). The UCR is the most comprehensive publicly available dataset on US crime, aggregating monthly reports from over 18,000 participating law-enforcement agencies.

The primary dependent variable in my analyses is larceny-theft, encompassing offenses such as shoplifting, pocket-picking, and bicycle theft, defined as the unlawful taking of property from the possession or constructive possession of another. Additional dependent variables include burglary and motor vehicle theft. Note that UCR reporting follows a "hierarchy rule": if multiple crimes are committed in a single incident, only the most serious offense is recorded. The hierarchy, from most serious to least serious, is homicide, rape, robbery, aggravated assault, burglary, theft (other than of a motor vehicle), motor vehicle theft, arson, and simple assault.

A well-documented limitation of the UCR data is distinguishing between true zeros and missing data. A recorded value of zero for a specific crime in a given month may indicate that no crimes occurred, or it may indicate that the agency simply failed to report (Kaplan 2023). Before aggregating the agency-year-month data to the county-year-quarter level, I restrict the sample to

law-enforcement agencies that report at least one crime (larceny-theft, burglary, motor vehicle theft, arson, or any violent crime) for all 12 months of every year they appear in the dataset.[7] Because my identification strategy relies on the relationship between *changes* in minimum wages and *changes* in local property crime rates, restricting the panel to consistently reporting agencies prevents artificial variation driven by reporting gaps. Moreover, my preferred specifications control directly for clearance rates, the number of police agencies reporting, and the number of violent crimes to account for unobserved changes in law enforcement practices, agency reporting behaviors, and broader county-level crime trends (Braun 2019).

Finally, UCR crime rates represent a lower bound on actual crime, as many property crimes go unreported to law enforcement (Raphael 2016). However, this under-reporting will not bias the estimates unless changes in the propensity to report crimes are systematically correlated with local minimum wage changes (conditional on the included controls).

3.2. Lightcast Vacancy Data

To measure local labor market concentration, I use online job vacancy data from Lightcast. This dataset contains the near-universe of online US job vacancies aggregated from approximately 50,000 websites, providing detailed information on occupation, location, and employer name.

Not all occupations are equally represented in online job postings (Azar et al. 2024; Hershbein and Kahn 2018). To maintain measurement credibility, I follow Azar et al. (2024) and restrict the concentration analysis to three specific minimum-wage-heavy retail occupations: Cashiers (SOC 41-2011), Retail Salespersons (SOC 41-2031), and Stock Clerks (SOC 43-5081).[8]

I calculate the Herfindahl–Hirschman Index (HHI) for each county r and occupation o at time t as follows:

$$\text{HHI}_{ort} = \sum_{j=1}^{N_{rt}} \left(\frac{q_{jort}}{\sum_{j=1}^{N_{rt}} q_{jort}} \right)^2, \tag{3}$$

where q_{jort} is employer j's number of vacancy postings for occupation o in county r at time t, and N_{rt} is the total number of hiring firms in that local market. Employers are identified by the

[7] I also drop two agencies due to clear data entry errors: "GA10800" (an agency in Oconee County, Georgia) and "AL00201" (an agency in Mobile County, Alabama). The results do not change if these agencies are included.

[8] As explained by Azar et al. (2024), job vacancies for the restaurant industry (Accommodation and Food Services) are not accurately reflected in online advertisements. See footnote 4.

standardized employer name attached to the vacancy. If a vacancy is missing an employer name, I conservatively assume it corresponds to a unique independent firm. This approach generally understates the true degree of employer concentration, both because missing names are likely posted by existing firms in the database, and because distinct subsidiary names may roll up to the same parent corporation. Furthermore, if large firms systematically hire multiple workers per single vacancy posting, this vacancy-based HHI will further understate true labor market concentration (Schubert et al. 2020).

This measure of labor market concentration relies on vacancy flows rather than employment stocks. However, prior validation using administrative data suggests that flow-based and stock-based HHI measures are highly positively correlated (Marinescu, Ouss, and Pape 2021).

3.3. Additional Variables

In my analyses, I define the effective minimum wage as the maximum of the federal and state minimum wage in a county. The main dependent variable for employment in my analyses is the total county-level employment in the retail sector (NAICS 452), obtained from the Quarterly Census of Employment and Wages (QCEW). See Appendix Table C.1 for an exhaustive list of data sources, including for all of the control variables in the analyses.

3.4. Summary Statistics

Table 1 presents the summary statistics for the final county-year-quarter estimation sample from 2010 to 2016. The primary explanatory variable, the prevailing effective minimum wage, averages $7.51 across the sample period. Importantly, there is substantial spatial and temporal variation in the minimum wage required for identification. As illustrated in Figure B.1, while the federal floor remained stagnant at $7.25, numerous states enacted staggered, independent wage increases, providing rich variation in the local wage floors.

To contextualize the economic magnitudes of the subsequent elasticity estimates, the middle panel of Table 1 reports the baseline levels for the dependent variables. The average county in the sample supports approximately 1,116 retail jobs. Within the property crime categories, larceny-theft is by far the most prevalent offense, averaging 601.1 incidents per county-quarter,

reflecting its nature as a high-frequency, low-barrier crime. Burglary and motor vehicle theft occur less frequently, averaging 190.8 and 77.3 incidents, respectively. Violent crimes are even less common, averaging 121.4 incidents per county-quarter.

Figure 1 plots the distribution of local labor market concentration—measured via the average HHI in each county across all years—for the three main retail occupations: Cashiers, Retail Salespersons, and Stock Clerks. The distributions confirm significant spatial heterogeneity in employer market power. While a large mass of counties operates in highly competitive labor markets, each occupation exhibits a significant right tail. A substantial fraction of local labor markets falls at or above the Department of Justice's threshold for severe concentration (HHI ≥ 0.25), providing the necessary empirical variation to evaluate the moderating role of labor market concentration on employment and property crime rates.

Finally, young adults are disproportionately represented both in minimum-wage jobs and in larceny-theft arrests, motivating studies of the effect of the minimum wage on crime (Braun 2019; Fone et al. 2023). Appendix Figure B.2 shows that the majority of individuals arrested for larceny-theft are young adults and that young adults comprise the majority of the workforce in cashier and retail sales occupations.

4. Empirical Strategy

The empirical strategy builds on the framework developed by Azar et al. (2024), augmented to account for the criminal justice and demographic covariates central to this context. Azar et al. (2024) demonstrate that the employment elasticity of the minimum wage in the retail sector is a positive function of local labor market concentration, specifically for three highly exposed occupations: stock clerks (SOC 43-5081), retail salespersons (SOC 41-2031), and cashiers (SOC 41-2011). To benchmark my data, I first replicate their core employment results. Building on this foundation, the primary contribution of this paper is to estimate how this local wage-setting power moderates the effect of the minimum wage on property crime.

To estimate the employment or crime elasticity of the minimum wage as a function of labor market concentration, I estimate the following panel fixed-effects specification separately for

each occupation o:

$$\log(Y_{rt}) = \alpha + \beta \log(\mathrm{MW}_{rt}) + \delta \left(\log(\mathrm{MW}_{rt}) \times \overline{\mathrm{HHI}}_{or} \right) + \gamma X_{rt} + \eta_r + \tau_{ct} + \lambda_s t + \varepsilon_{rt}, \quad (4)$$

where Y_{rt} is the outcome variable of interest (e.g., retail employment or property crime) in county r and quarter t.[9] The variable MW_{rt} is the prevailing effective minimum wage (the maximum of the federal or state rate), and $\overline{\mathrm{HHI}}_{or} \equiv \frac{1}{28} \sum_t \mathrm{HHI}_{ort}$ is the average Herfindahl-Hirschman Index for retail occupation o in county r. The specification includes county fixed effects (η_r) to control for time-invariant local labor market characteristics, Census-region-by-year-quarter fixed effects (τ_{ct}) to absorb regional macroeconomic shocks, and state-specific linear time trends ($\lambda_s t$) to account for unobserved, smoothly evolving state-level factors.[10] I also estimate a second specification replacing the continuous concentration measure with an indicator for highly concentrated markets:

$$\log(Y_{rt}) = \alpha + \beta \log(\mathrm{MW}_{rt}) + \delta \left(\log(\mathrm{MW}_{rt}) \times \mathbb{1}[\overline{\mathrm{HHI}}_{or} \geq 0.25] \right) + \gamma X_{rt} + \eta_r + \tau_{ct} + \lambda_s t + \varepsilon_{rt}, \quad (5)$$

The threshold of 0.25 is explicitly chosen to align with the US Department of Justice (DOJ) guidelines, which classify markets with an HHI at or above 0.25 (or 2,500 on a 10,000-point scale) as "highly concentrated." Standard errors in all specifications are clustered at the state level to account for serial correlation and the state-level variation in minimum wage policy.

Following Azar et al. (2024), I use the time-averaged Herfindahl-Hirschman Index, $\overline{\mathrm{HHI}}_{or}$, rather than the time-varying measure, HHI_{ort}, to measure local labor market concentration. The time-varying HHI_{ort} introduces potential endogeneity: minimum wage increases can force smaller, less productive firms to exit, which mechanically increases concentration while simultaneously destroying jobs. Using the time-averaged $\overline{\mathrm{HHI}}_{or}$ helps mitigate this concern. To the extent that residual endogeneity remains, it biases the estimated employment elasticity toward zero, making any captured positive employment and crime reduction effects inherently conservative. An alternative approach is to use a pre-determined HHI measure calculated from

[9] Retail sector employment consists of all workers in NAICS 452 from the QCEW.

[10] Because $\overline{\mathrm{HHI}}_{or}$ is time-invariant, its direct effect is perfectly collinear with the county fixed effects. Therefore, only the interaction term is included in the regression.

an initial period (2010 to 2011) and restrict the analysis to subsequent years (2012 to 2016). I implement this strategy in my robustness checks and find highly consistent results.

The vector of time-varying covariates, X_{rt}, contains both economic and criminal justice controls. Following Azar et al. (2024), the economic controls consist of the log of total population, the log of total employment, the log of total earnings, and the log of the unemployment rate.[11] These variables control for local demand shocks, shifting productivity (proxied by total earnings), and broader business cycle fluctuations that may be correlated with both the timing of minimum wage hikes and criminal activity.

Crucially, because measured crime rates are sensitive to changes in law enforcement behavior, X_{rt} also includes a rich set of criminal justice controls following Braun (2019). I control for the log of violent crime, the number of police agencies reporting, and the log of clearance rates for both property and violent crimes. Clearance rates—defined as the ratio of arrests to reported crimes—serve as a proxy for unobserved shifts in policing intensity and the probability of apprehension. Controlling for violent crime and the number of reporting agencies helps isolate the property crime effect from general trends in criminality or administrative reporting changes.

Finally, as a robustness check, I estimate models that include demographic controls to account for shifts in the composition of the local population. These include the fraction of the population that is male, the fraction that is white, and the fraction of the population falling into discrete age bins (16–24, 25–34, 35–44, 45–54, 55–64, and 65+).

5. Results

5.1. The First Stage: Employment Effects

Before examining the spillover effects of the minimum wage on property crime, I first establish the direct effect on the legal labor market. In the context of the theoretical framework, this serves as the "first stage" of the analysis: if employer market power mitigates the displacement effects of a wage floor, we should observe the employment elasticity of the minimum wage becoming

[11] The left-hand side variable in the main specification is employment in the *retail sector*, whereas the employment and earnings variables included as controls on the right-hand side are aggregate variables, i.e., the totals across *all* industries.

more positive as local labor market concentration increases.

Figure 2 presents the results of the continuous specification (Equation (4)), which interacts the log of the minimum wage with the continuous concentration index. Consistent with Azar et al. (2024), I find a strong, positive gradient between concentration and the employment elasticity. In highly competitive markets ($\overline{\text{HHI}}_{or} \approx 0$), the baseline coefficient on the minimum wage is negative, reflecting standard competitive displacement effects. However, the interaction term is large, positive, and statistically significant at the 5% level across all three retail occupations (Panel a). Exact point estimates and standard errors are available in Appendix Table C.2.

To quantify the magnitude of this effect in highly concentrated markets, I turn to the discrete specification (Equation (5)), replacing the continuous concentration index with an indicator for $\overline{\text{HHI}}_{or} \geq 0.25$. The results are presented in Figure 3, with exact point estimates and standard errors in Appendix Table C.3. In highly concentrated labor markets, the minimum wage *increases* employment: summing the baseline and interaction coefficients across the three targeted retail occupations yields an average net employment elasticity of 0.16. Evaluated at the sample means, a \$0.10 increase in the minimum wage—representing a roughly 1.33% increase from the average baseline of \$7.51—translates to a 0.21% increase in retail employment. For the average county in the sample, this equates to the addition of approximately 2.34 retail jobs. This simultaneous increase in both the wage rate and employment opportunities provides the necessary mechanism for an unambiguous reduction in local criminal participation.

5.2. Main Results: Property Crime

Having established that wage floors increase employment in concentrated labor markets, I next evaluate the downstream effects on local property crime rates. Because property crimes vary in their responsiveness to low-wage labor market conditions, I estimate the model separately for larceny-theft, motor vehicle theft, and burglary.

The results for larceny-theft strongly support the theoretical prediction (Hypothesis 1). As shown in Panel b of Figure 2, the coefficient of the interaction term between the log of the minimum wage and the continuous concentration index (Equation (4)) is negative and statistically significant at the 5% level. This indicates that as employer concentration rises, the minimum

wage becomes an increasingly powerful theft reduction tool. Exact point estimates and standard errors are available in Appendix Table C.2.

Examining the results for the specification with the discrete indicator for high labor market concentration (Equation (5)), Figure 3 shows that the minimum wage results in a meaningful reduction in crime when labor markets are highly concentrated. Summing the baseline and interaction coefficients across the three occupations yields an average net larceny-theft elasticity of -0.31. These results imply that a $0.10 increase in the minimum wage is associated with a 0.41% reduction in larceny-theft. For the average county in the sample, this corresponds to approximately 2.46 fewer larceny-theft incidents per county-year-quarter. Appendix Table C.3 shows the exact point estimates and standard errors.

Relating this reduction in crime to the employment gains calculated in the previous subsection implies that for every new retail job created by the minimum wage hike, larceny-theft incidents fall by roughly 1.05. Because the minimum wage simultaneously increases both the employment level and the wage rate for existing workers, this ratio effectively serves as an upper bound on the pure employment (or extensive margin substitution) effect.

Turning to the other property crimes in Figure 2, I do not find evidence that the minimum wage reduces motor vehicle theft (Panel c) or burglary (Panel d) when labor markets are concentrated. While the point estimates for the interaction terms are generally negative, they are noisy and fail to reach conventional levels of statistical significance. This heterogeneity across crime types is consistent with the theoretical mechanism: larceny-theft (e.g., shoplifting) is a low-barrier, high-frequency offense most likely to serve as a substitute for low-wage employment. Burglary and motor vehicle theft, which often require more specialized skills, coordination, or capital, are expected to be less elastic to marginal changes in legal labor market conditions.

5.3. Placebo Tests

A primary threat to identification in a two-way fixed effects panel design is the presence of unobserved, time-varying local shocks. If counties that experience minimum wage hikes systematically alter their policing intensity or crime reporting practices, the main estimates could be spuriously driven by these institutional changes rather than by shifts in labor market

opportunities.

To directly address this concern, I conduct a falsification test using violent crime as the dependent variable. Note that since the log of violent crime is the dependent variable in these regressions, I estimate Equations (4) and (5) *without* the log of violent crime on the right-hand side (although I continue to include all other criminal justice controls in the specification). Because violent crimes (such as aggravated assault) are predominantly driven by interpersonal conflict and socio-emotional factors rather than pure economic substitution, they should be theoretically unresponsive to marginal fluctuations in low-wage labor market conditions.

Figure 4 reports the results of these placebo regressions. Across both the continuous and discrete specifications, the interaction terms for violent crime are statistically indistinguishable from zero (exact point estimates are provided in Appendix Table C.4). Furthermore, the point estimates are often positive, contrasting sharply with the robust negative elasticities observed for larceny-theft. This placebo test provides reassuring evidence that the property crime reductions documented in the main results are not artifacts of unobservable local trends in law enforcement or broader crime reporting behaviors.

5.4. Robustness Checks

To test the sensitivity of the main results, I estimate several alternative specifications. The results of these robustness exercises are presented in Figure 5 (Continuous HHI) and Figure 6 (Discrete HHI), with the corresponding regression tables available in the Appendix.

First, one might be concerned that the inclusion of criminal justice controls (clearance rates, violent crime, and the number of reporting police agencies) introduces post-treatment bias. Estimating the model without these crime controls leaves the point estimates virtually unchanged, confirming that the main effects are driven by labor market dynamics, not endogenous shifts in the control variables. Second, I saturate the model with a rich set of demographic controls—including county-level age, sex, and racial distributions—to account for shifting population demographics. The inclusion of these covariates does not meaningfully alter the coefficients.

Finally, there is a potential concern regarding the endogeneity of the concentration measure

itself, as concurrent local economic shocks could theoretically influence both the minimum wage's bite and employer consolidation. To alleviate this, I construct a predetermined measure of local labor market concentration using Lightcast vacancy data strictly from 2010 to 2011. I then re-estimate the main regressions using only the subsequent 2012–2016 panel to completely avoid temporal overlap. Relying exclusively on this predetermined HHI yields results that are both qualitatively and quantitatively consistent with the baseline findings.

6. Conclusion

This paper bridges two prominent strands of empirical literature—the economics of crime and the growing evidence of employer market power—to investigate how local labor market concentration moderates the relationship between the minimum wage and property crime. Traditional Beckerian models of crime often imply a stark trade-off when evaluating wage floors: while higher wages increase the opportunity cost of illicit activity for employed individuals, the resulting job displacement may push marginalized workers toward economically motivated crimes. By integrating criminal participation into an oligopsony framework, I demonstrate that when employers wield significant market power, this conventional trade-off no longer applies and the effect of the minimum wage on property crime is unambiguously negative.

Using a novel county-year-quarter panel spanning 2010 to 2016 that links FBI Uniform Crime Reporting data with near-universe online job vacancy records from Lightcast, I find empirical support for this theoretical prediction. In highly concentrated retail labor markets (HHI ≥ 0.25), minimum wage increases do not induce standard competitive disemployment effects. Instead, they significantly boost both employment and the prevailing wage. This increase in legal labor market opportunities generates a substantial decline in larceny-theft.

Importantly, the findings reveal that this crime reduction effect is concentrated among low-barrier, high-frequency offenses. While larceny-theft is sensitive to local low-wage labor market conditions, more specialized or capital-intensive offenses like burglary and motor vehicle theft remain largely unresponsive. Falsification tests confirming a null effect on violent crime provide evidence that these reductions in property crime are driven by shifting labor market dynamics rather than unobserved local shocks to law enforcement practices or crime reporting.

These results have meaningful implications for the design and evaluation of low-wage labor market policies. They suggest that the social returns to minimum wage increases are systematically underestimated if positive spillovers onto local public safety are ignored. Furthermore, the findings highlight the critical importance of local market structure in determining policy efficacy. In an era characterized by rising corporate consolidation and pervasive monopsony power in the low-wage sector, wage floors can serve as an effective, broad-based policy tool not only for raising the living standards of vulnerable workers, but also for meaningfully reducing local property crime.

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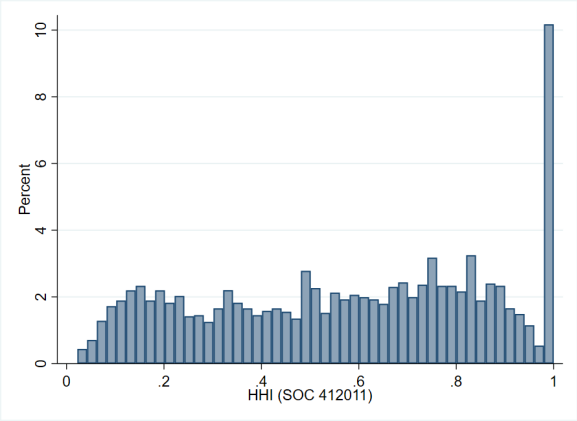
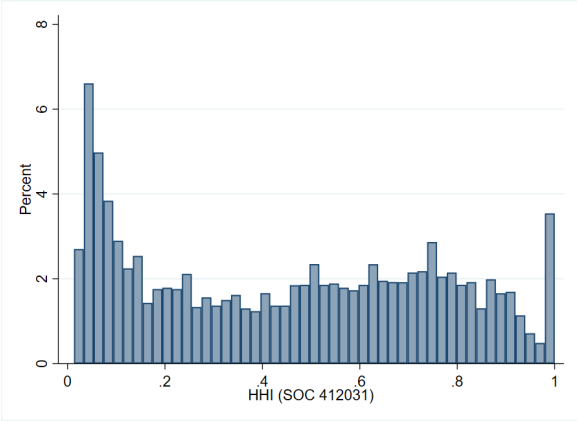
Figures

Figure 1. Distribution of Average Labor Market Concentration ($\overline{\text{HHI}}$) by Retail Occupation

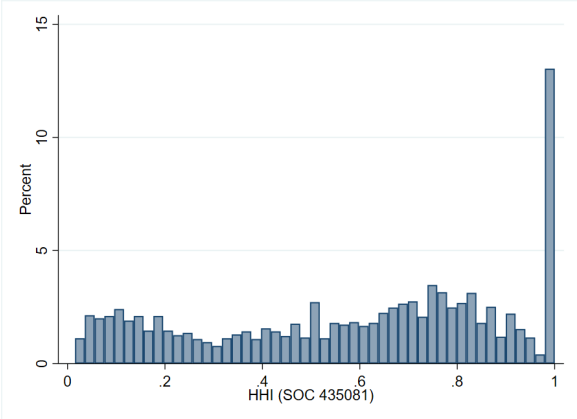
(a) Cashiers (SOC 41-2011)



(b) Retail Sales (SOC 41-2031)

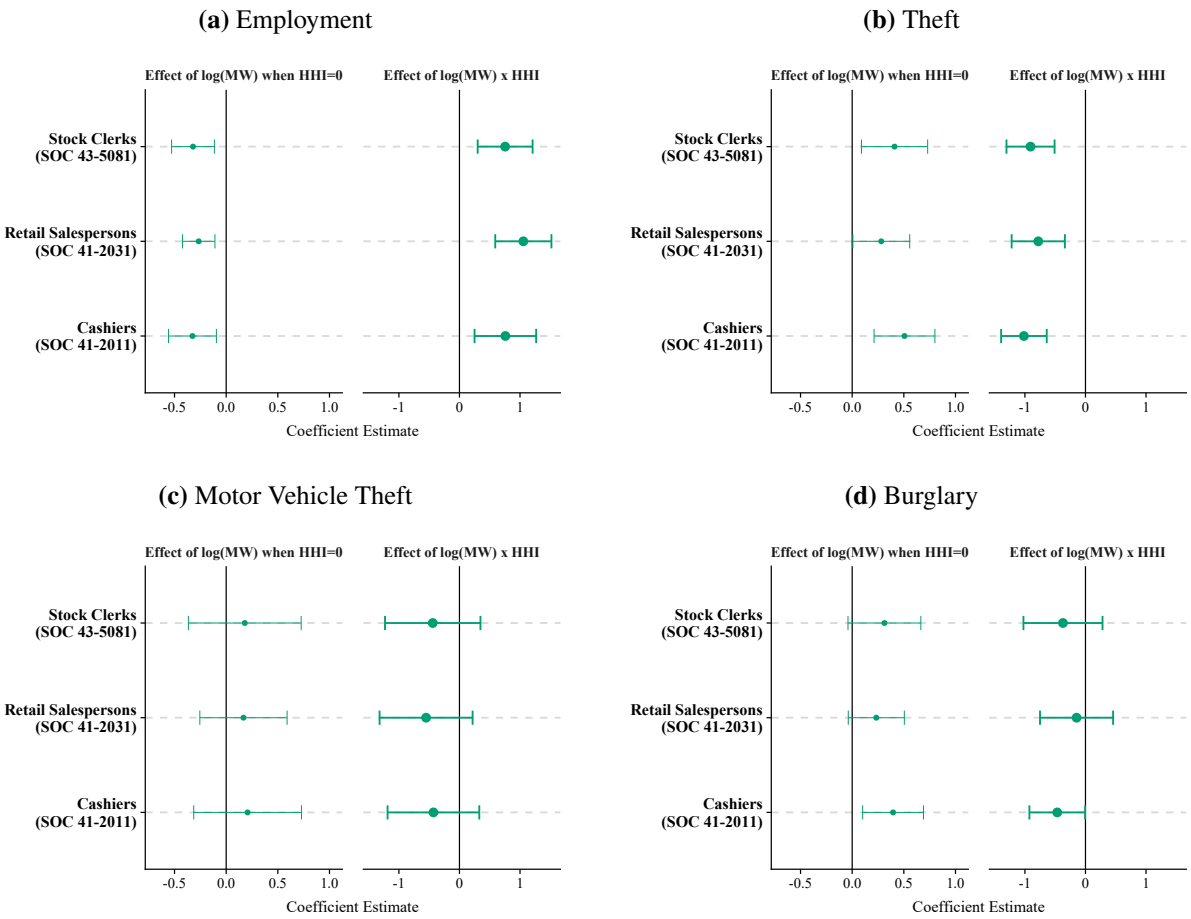


(c) Stock Clerks (SOC 43-5081)



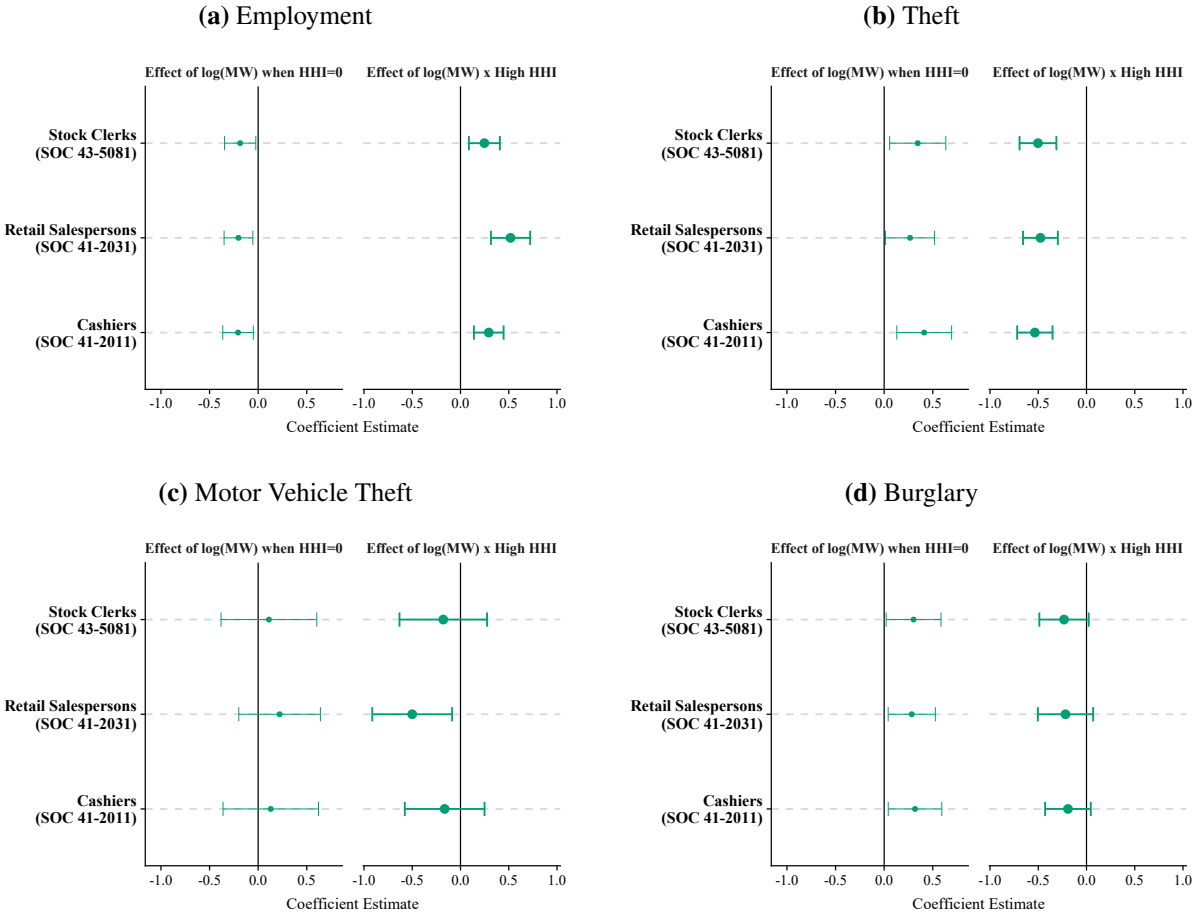
Note: This figure illustrates the spatial variation in local labor market concentration across US counties for three minimum-wage-heavy retail occupations: Cashiers, Retail Salespersons, and Stock Clerks. The concentration measure plotted is the average Herfindahl-Hirschman Index (HHI) of each county and occupation over the sample period. While many local markets are highly competitive, the distributions reveal a meaningful right tail of counties exhibiting significant employer concentration. *Source:* Author's calculations using the online job vacancy data from Lightcast.

Figure 2. Minimum Wage Elasticities of Employment and Property Crime by Continuous Labor Market Concentration



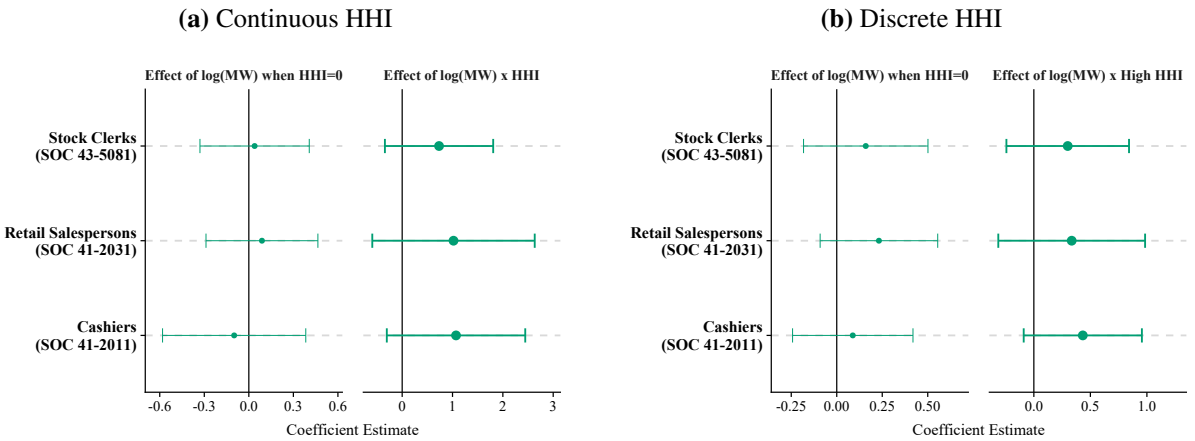
Note: This figure presents estimates of the elasticity of employment and property crime with respect to the minimum wage as a function of local labor market concentration. Each panel plots both the baseline log(MW) coefficient and its interaction with the continuous concentration index ($\overline{\text{HHI}}$). The baseline estimates are obtained from estimating Equation (4). The dependent variables vary by panel: log retail employment (panel a), log larceny-theft (panel b), log motor vehicle theft (panel c), and log burglary (panel d). The data are a county-year-quarter panel spanning 2010–2016. Local concentration is defined using the time-averaged HHI for Stock Clerks, Retail Salespersons, and Cashiers. All specifications include county fixed effects, census-region-by-year-quarter fixed effects, and state-specific linear time trends. Time-varying economic controls include the log of county population, total aggregate employment, total earnings, and the unemployment rate. Criminal justice controls include log violent crime, log clearance rates for property and violent crimes, and the number of reporting police agencies. Error bars represent 95% confidence intervals, with standard errors clustered at the state level.

Figure 3. Minimum Wage Elasticities of Employment and Property Crime in Highly Concentrated Labor Markets



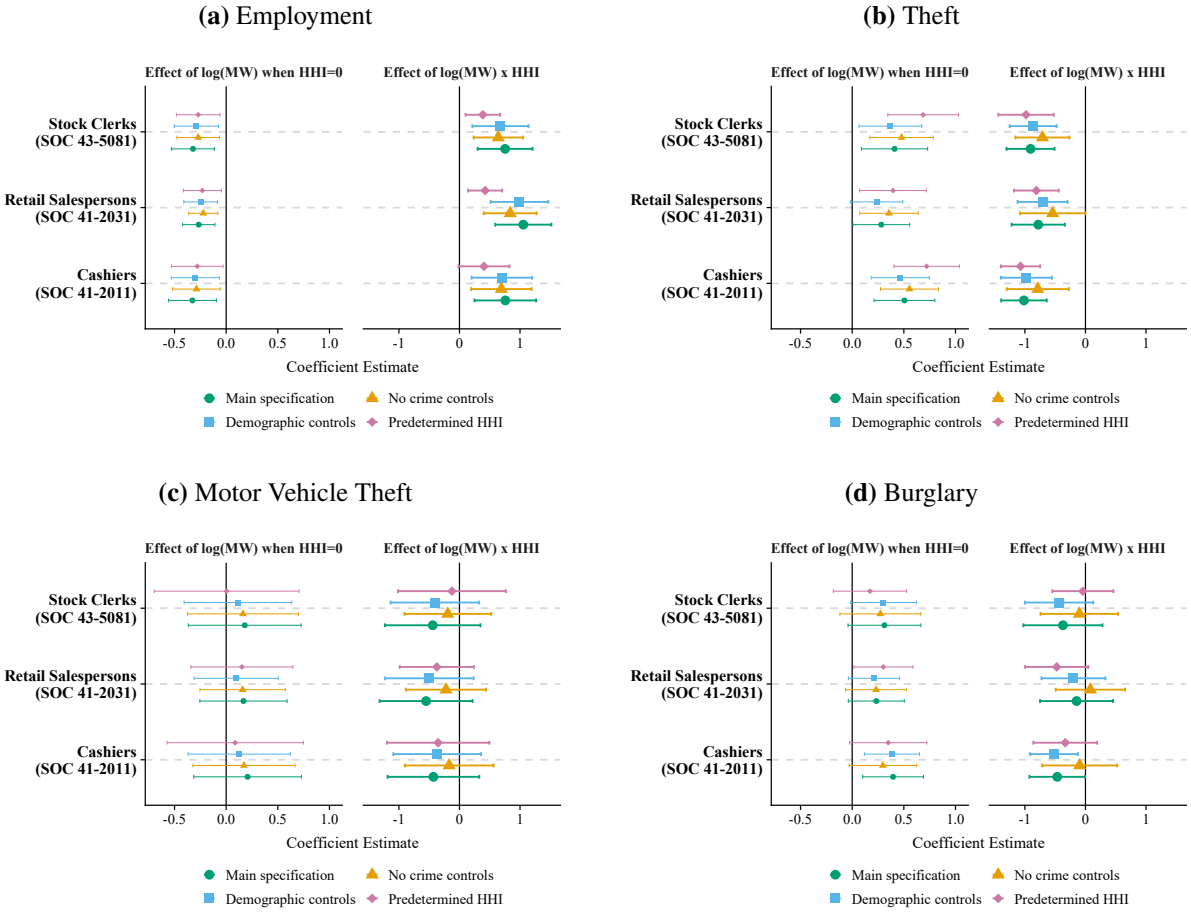
Note: This figure presents estimates of the elasticity of employment and property crime with respect to the minimum wage as a function of local labor market concentration. Each panel plots both the baseline log(MW) coefficient and its interaction with the discrete indicator for a highly concentrated local labor market (High HHI, defined as $\overline{\text{HHI}} \geq 0.25$). The baseline estimates are obtained from estimating Equation (5). The dependent variables vary by panel: log retail employment (panel a), log larceny-theft (panel b), log motor vehicle theft (panel c), and log burglary (panel d). The data are a county-year-quarter panel spanning 2010–2016. Local concentration is defined using the time-averaged HHI for Stock Clerks, Retail Salespersons, and Cashiers. All specifications include county fixed effects, census-region-by-year-quarter fixed effects, and state-specific linear time trends. Time-varying economic controls include the log of county population, total aggregate employment, total earnings, and the unemployment rate. Criminal justice controls include log violent crime, log clearance rates for property and violent crimes, and the number of reporting police agencies. Error bars represent 95% confidence intervals, with standard errors clustered at the state level.

Figure 4. Placebo Tests: Minimum Wage Elasticities of Violent Crime by Labor Market Concentration



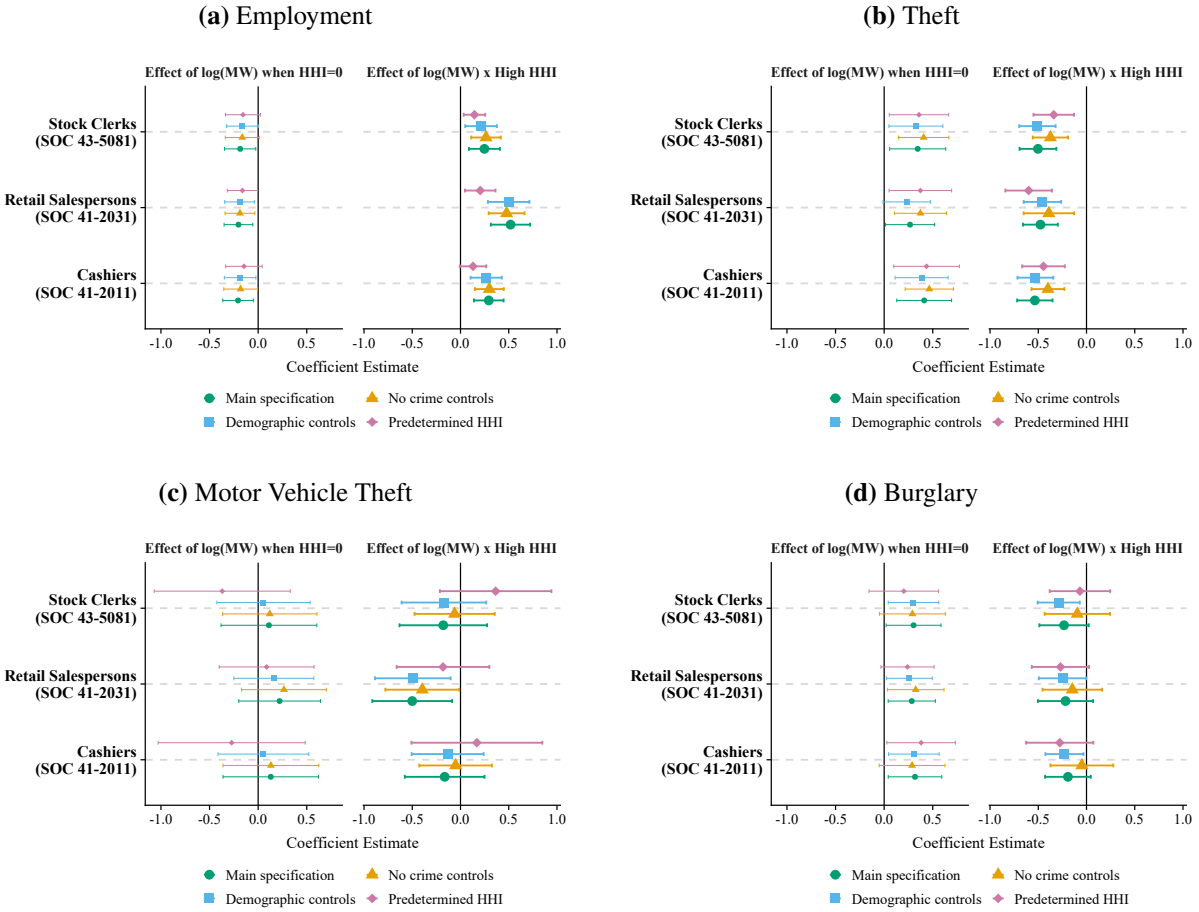
Note: This figure presents placebo test estimates of the elasticity of violent crime with respect to the minimum wage, alongside its interaction with local labor market concentration. Panel a interacts the minimum wage with the continuous measure of local concentration ($\overline{\text{HHI}}$) using Equation (4). Panel b utilizes the discrete indicator (High HHI, defined as $\overline{\text{HHI}} \geq 0.25$) using Equation (5). The dependent variable is log violent crime. The data are a county-quarter panel spanning 2010–2016. Local concentration is defined using the time-averaged HHI for Stock Clerks, Retail Salespersons, and Cashiers. All specifications include county fixed effects, census-region-by-year-quarter fixed effects, and state-specific linear time trends. Time-varying economic controls include the log of county population, total aggregate employment, total earnings, and the unemployment rate. Criminal justice controls include log clearance rates for property and violent crimes, and the number of reporting police agencies; log violent crime is excluded as a control here, as it serves as the dependent variable. Error bars represent 95% confidence intervals, with standard errors clustered at the state level.

Figure 5. Robustness Checks: Minimum Wage Elasticities of Employment and Property Crime by Continuous Labor Market Concentration



Note: This figure presents estimates of the elasticity of employment and property crime with respect to the minimum wage as a function of local labor market concentration. Each panel plots both the baseline log(MW) coefficient and its interaction with the continuous concentration index ($\overline{\text{HHI}}$). The baseline estimates are obtained from estimating Equation (4). The dependent variables vary by panel: log retail employment (panel a), log larceny-theft (panel b), log motor vehicle theft (panel c), and log burglary (panel d). The data are a county-year-quarter panel spanning 2010–2016. Local concentration is defined using the time-averaged HHI for Stock Clerks, Retail Salespersons, and Cashiers. Estimates are shown for the main specification alongside various robustness checks: excluding crime controls, adding demographic controls, and using a predetermined HHI (calculated using 2010–2011 vacancy data, with the regression estimated on a 2012–2016 panel to avoid overlap). Base specifications include county fixed effects, census-region-by-year-quarter fixed effects, state-specific linear time trends, and time-varying economic controls. Error bars represent 95% confidence intervals, with standard errors clustered at the state level.

Figure 6. Robustness Checks: Minimum Wage Elasticities of Employment and Property Crime in Highly Concentrated Labor Markets



Note: This figure presents estimates of the elasticity of employment and property crime with respect to the minimum wage as a function of local labor market concentration. Each panel plots both the baseline log(MW) coefficient and its interaction with the discrete indicator for a highly concentrated local labor market (High HHI, defined as $\overline{\text{HHI}} \geq 0.25$). The baseline estimates are obtained from estimating Equation (5). The dependent variables vary by panel: log retail employment (panel a), log larceny-theft (panel b), log motor vehicle theft (panel c), and log burglary (panel d). The data are a county-year-quarter panel spanning 2010–2016. Local concentration is defined using the time-averaged HHI for Stock Clerks, Retail Salespersons, and Cashiers. Estimates are shown for the main specification alongside various robustness checks: excluding crime controls, adding demographic controls, and using a predetermined HHI (calculated using 2010–2011 vacancy data, with the regression estimated on a 2012–2016 panel to avoid overlap). Base specifications include county fixed effects, census-region-by-year-quarter fixed effects, state-specific linear time trends, and time-varying economic controls. Error bars represent 95% confidence intervals, with standard errors clustered at the state level.

Tables

Table 1. Summary Statistics

Variable	Mean	SD
Labor Market & Economic Variables		
Minimum Wage	7.51	0.51
Avg HHI (Cashiers)	0.56	0.29
Avg HHI (Stock Clerks)	0.57	0.31
Avg HHI (Retail Sales)	0.44	0.31
Retail Employment	1,116	3,038
Total Earnings (millions)	474.6	2,078
Total Employment (thousands)	50.4	155.8
Population (thousands)	110.0	338.5
Unemployment Rate	7.19	2.96
Crime & Criminal Justice Variables		
Larceny-Theft	601.1	1,811
Burglary	190.8	597.1
Motor Vehicle Theft	77.28	379.9
Violent Crime	121.4	488.8
Property Clearance Rate	0.22	0.14
Violent Clearance Rate	0.54	0.28
Demographic Controls		
Fraction White	0.86	0.16
Fraction Male	0.50	0.02
Fraction Aged 15–24	0.13	0.03
Fraction Aged 25–34	0.12	0.02
Fraction Aged 35–44	0.12	0.02
Fraction Aged 45–54	0.14	0.02
Fraction Aged 55–64	0.14	0.02
Fraction Aged 65+	0.17	0.05

Note: This table presents summary statistics for the county-year-quarter panel spanning 2010–2016. See Section 4 for a discussion of the empirical specification and how these variables are used in the analyses. Minimum Wage represents the effective minimum wage (the maximum of the federal and state rates) prevailing in a given county. Avg HHI ($\overline{\text{HHI}}_{or}$) measures local employer concentration for a specific retail occupation. It is defined as the time-averaged Herfindahl–Hirschman Index, constructed from online job vacancy shares using Lightcast data. Retail Employment is the total retail sector (NAICS 452) employment level sourced from the QCEW. Crime variables include total county-year-quarter incident counts compiled from the FBI's UCR Program (Kaplan 2019). Clearance rates proxy for local policing intensity and are defined as the ratio of total arrests to total reported offenses. Additional economic and demographic control variables are sourced from the BLS (LAUS and QCEW) and SEER databases, respectively, and are incorporated into the primary empirical specifications.

Online Appendix for "Labor Market Concentration, Minimum Wages, and Local Property Crime Rates"

Appendix A. Mathematical Proofs

A.1. Proof of Proposition 1: Wage-Taking Regime

In a perfectly competitive labor market, the introduction of a minimum wage $\bar{w}$ above the market-clearing wage w^* induces rationing, as labor supply $L(\bar{w})$ strictly exceeds labor demand $D(\bar{w})$. The mass of rationed workers is given by $R(\bar{w}) = L(\bar{w}) - D(\bar{w})$.

Assume that employment is rationed randomly among those willing to work. A fraction of the rationed workers will select property crime as their second-best option. A worker prefers crime to non-participation if $r_i > b_i$. Conditional on being in the labor force (i.e., $w \geq r_i$ and $w \geq b_i$), the probability of selecting crime when rationed is:

$$\begin{aligned}\Omega(\bar{w}) &= \mathbb{P}(r_i > b_i \mid \bar{w} \geq r_i \text{ and } \bar{w} \geq b_i) \\ &= \frac{\mathbb{P}(\bar{w} \geq r_i > b_i)}{\mathbb{P}(\bar{w} \geq r_i \text{ and } \bar{w} \geq b_i)} \\ &= \frac{\int_{-\infty}^{\bar{w}} \int_{b}^{\bar{w}} g(r, b)\, dr\, db}{G(\bar{w}, \bar{w})}\end{aligned} \tag{6}$$

Total property crime C_{total} is the sum of unconstrained crime and crime committed by rationed workers:

$$C_{total}(\bar{w}) = C(\bar{w}) + \Omega(\bar{w}) \cdot [L(\bar{w}) - D(\bar{w})] \tag{7}$$

Differentiating with respect to $\bar{w}$ yields:

$$\frac{\partial C_{total}}{\partial \bar{w}} = C'(\bar{w}) + \Omega'(\bar{w})[L(\bar{w}) - D(\bar{w})] + \Omega(\bar{w})[L'(\bar{w}) - D'(\bar{w})] \tag{8}$$

Evaluating this derivative at the initial market-clearing wage $\bar{w} = w^*$, the rationing term

$[L(w^*) - D(w^*)]$ is exactly zero. The expression thus simplifies to:

$$\left.\frac{\partial C_{total}}{\partial \bar{w}}\right|_{\bar{w}=w^*} = C'(w^*) + \Omega(w^*) \cdot [L'(w^*) - D'(w^*)] \tag{9}$$

This completes the proof. □

A.2. Proof of Proposition 2: Oligopsony Regime

Assume N firms engage in Cournot competition. Firm j maximizes profit $\pi_j = f(l_j) - w(L)l_j$, where $w(L)$ is the inverse of the joint supply function $L(w) = G(w, w)$. The first-order condition for firm j is $f'(l_j) = w(L) + l_j w'(L)$.

In a symmetric equilibrium where $l_j = L/N$, the marginal revenue product of labor strictly exceeds the wage: $f'(L/N) > w^*$. This gap is the markdown wedge.

If a minimum wage $\bar{w}$ is imposed such that $w^* < \bar{w} \leq f'(L/N)$, the marginal cost of labor becomes perfectly elastic at $\bar{w}$ up to the new quantity supplied $L(\bar{w})$. Because $\bar{w}$ does not exceed the marginal revenue product of labor, firms are willing to hire all workers supplied at $\bar{w}$. Therefore, $D(\bar{w}) = L(\bar{w})$, and no rationing occurs ($R(\bar{w}) = 0$).

Since no workers are rationed, total crime is simply given by the unconstrained supply of crime evaluated at $\bar{w}$:

$$C_{total}(\bar{w}) = C(\bar{w}) = \int_{\bar{w}}^{\infty} \left(\int_{-\infty}^{r} g(r, b)\, db \right) dr \tag{10}$$

Differentiating with respect to $\bar{w}$ using Leibniz's rule gives:

$$\frac{\partial C_{total}}{\partial \bar{w}} = -\int_{-\infty}^{\bar{w}} g(\bar{w}, b)\, db < 0 \tag{11}$$

This opportunity cost effect is strictly negative since $f(r, b)$ is a non-negative density. □

A.3. Proof of Proposition 3: Concentration and Crime Elasticity

Let labor market concentration be defined by the Herfindahl-Hirschman Index (HHI), which in a symmetric N-firm oligopoly simplifies to $H = N^{-1}$.

The symmetric Cournot equilibrium wage w^* satisfies:

$$f'(L/N) = w^* \left(1 + \frac{H}{\varepsilon_{L,w}}\right) \tag{12}$$

where $\varepsilon_{L,w}$ is the elasticity of legal labor supply. Holding the production function and labor supply constant, an increase in concentration H strictly decreases the equilibrium wage w^* ($\frac{\partial w^*}{\partial H} < 0$).

From Proposition 2, the marginal effect of the minimum wage in the non-rationing region evaluated at the initial equilibrium is $\frac{\partial C}{\partial \bar{w}}\Big|_{\bar{w}=w^*} = C'(w^*)$. We want to find how this marginal effect changes with concentration:

$$\frac{\partial}{\partial H}\left(C'(w^*)\right) = C''(w^*) \cdot \frac{\partial w^*}{\partial H} \tag{13}$$

Assume the density of marginal criminals is decreasing at the prevailing wage, such that the crime supply curve is convex ($C''(w) > 0$). This is a standard regularity condition satisfied if the distribution of illegal returns is right-skewed.

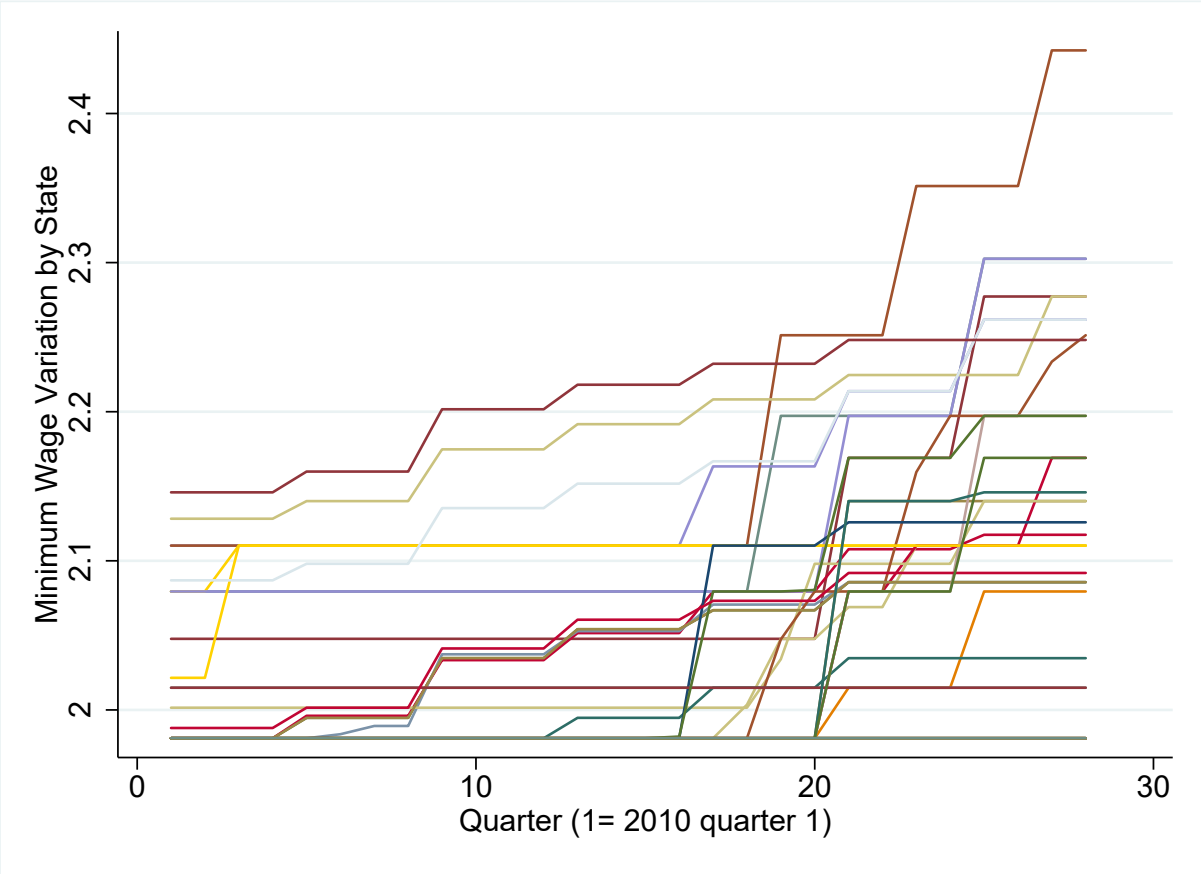
Since $C''(w^*) > 0$ and $\frac{\partial w^*}{\partial H} < 0$, it follows that:

$$\frac{\partial^2 C}{\partial \bar{w} \partial H} < 0 \tag{14}$$

This indicates that the crime-reducing effect of the minimum wage becomes larger in magnitude (more negative) as local labor market concentration increases. □

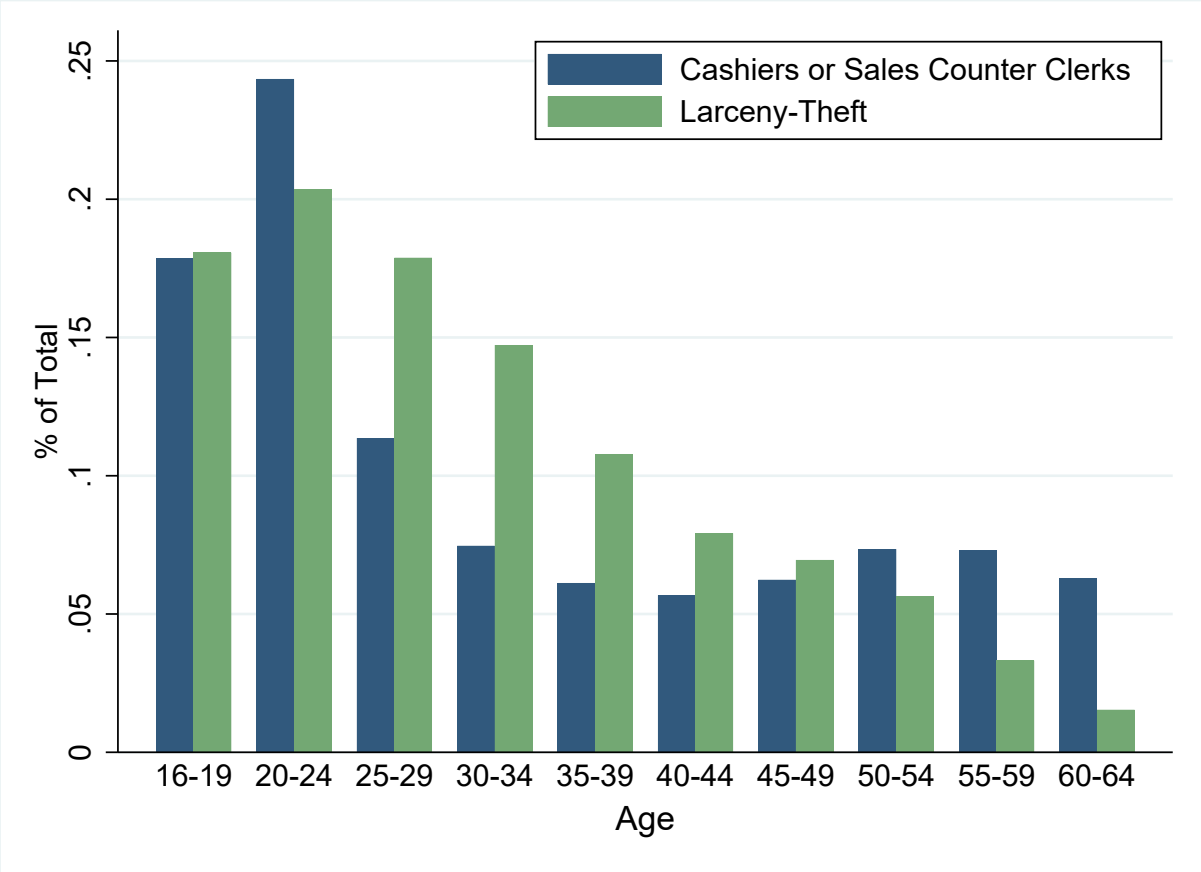
Appendix B. Figures

Figure B.1. Minimum wage changes over sample period



Note: This figure plots the temporal and spatial variation in the prevailing effective minimum wage from 2010 Q1 to 2016 Q4. The effective minimum wage is defined as the maximum of the federal or state minimum wage in each county. Each line in the graph corresponds to one state in the data. This temporal and spatial variation provides the identifying variation for the empirical strategy.

Figure B.2. Comparison of the Age Distributions of Workers in Retail Occupations and Arrests for Larceny-Theft



Note: This figure compares the age distribution of individuals employed in selected retail occupations with the age distribution of individuals arrested for larceny-theft. Plotted in blue are the fraction of workers in occupations OCC275 (Sales Counter Clerks) or OCC276 (Cashiers) in each age category. Plotted in green are the fraction of arrests for larceny-theft in each age category. All data are for the year 2015. Occupation data come from the American Community Survey (ACS) and crime data come from the FBI's Uniform Crime Reporting (UCR) Program Kaplan (2019).

Appendix C. Tables

Table C.1. Data sources and variable descriptions

Variable	Source
Minimum wage	Allegretto et al. (2017)
Minimum wage (2015–2016)	Government sources
Market concentration (HHI)	Lightcast
Larceny-theft	Kaplan (2019)
Employment (NAICS 452)	QCEW
Unemployment rate	LAUS
Total earnings	QCEW
Total employed	LAUS
Clearance rates	Kaplan (2019)
Violent crime rate	Kaplan (2019)
Demographic variables	NBER [originally SEER]

Note: This table outlines the primary variables used in the empirical analysis and their corresponding data sources.

Table C.2. Minimum Wage Elasticities of Employment and Property Crime by Continuous Labor Market Concentration

VARIABLES	(1) Stock Clerks	(2) Retail Sales	(3) Cashiers
Panel A: Log Retail Employment			
log(MW)	-0.321***	-0.266***	-0.326***
	(0.103)	(0.0783)	(0.115)
log(MW) × $\overline{\text{HHI}}$	0.754***	1.055***	0.758***
	(0.226)	(0.231)	(0.253)
Observations	44,320	44,493	44,368
Panel B: Log Larceny-Theft			
log(MW)	0.338***	0.227*	0.462***
	(0.123)	(0.129)	(0.163)
log(MW) × $\overline{\text{HHI}}$	-0.988***	-1.056***	-1.149***
	(0.229)	(0.254)	(0.201)
Observations	59,035	59,929	59,121
Panel C: Log Motor Vehicle Theft			
log(MW)	0.181	0.168	0.209
	(0.272)	(0.210)	(0.259)
log(MW) × $\overline{\text{HHI}}$	-0.443	-0.553	-0.432
	(0.392)	(0.382)	(0.377)
Observations	51,862	52,361	51,933
Panel D: Log Burglary			
log(MW)	0.312*	0.233*	0.395***
	(0.176)	(0.135)	(0.147)
log(MW) × $\overline{\text{HHI}}$	-0.370	-0.146	-0.465**
	(0.325)	(0.300)	(0.229)
Observations	55,028	55,626	55,107
County & Time FE	✓	✓	✓
State-specific time trends	✓	✓	✓
Baseline controls	✓	✓	✓
Crime controls	✓	✓	✓

Note: This table presents estimates of the elasticity of employment and property crime with respect to the minimum wage as a function of local labor market concentration ($\overline{\text{HHI}}$), obtained from estimating Equation (4). The dependent variables vary by panel: log retail employment (Panel A), log larceny-theft (Panel B), log motor vehicle theft (Panel C), and log burglary (Panel D). The data are a county-quarter panel spanning 2010–2016. All specifications include county fixed effects, Census-region-by-year-quarter fixed effects, and state-specific linear time trends. Time-varying economic controls include the log of county population, total aggregate employment, total earnings, and the unemployment rate. Criminal justice controls include log violent crime, log clearance rates for property and violent crimes, and the number of reporting police agencies. Standard errors, reported in parentheses, are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.3. Minimum Wage Elasticities of Employment and Property Crime in Highly Concentrated Labor Markets

VARIABLES	(1) Stock Clerks	(2) Retail Sales	(3) Cashiers
Panel A: Log Retail Employment			
log(MW)	-0.185**	-0.203***	-0.207**
	(0.0800)	(0.0736)	(0.0783)
log(MW) × High $\overline{\text{HHI}}$	0.248***	0.518***	0.293***
	(0.0799)	(0.101)	(0.0765)
Observations	44,320	44,493	44,368
Panel B: Log Larceny-Theft			
log(MW)	0.249**	0.188*	0.317**
	(0.118)	(0.112)	(0.127)
log(MW) × High $\overline{\text{HHI}}$	-0.530***	-0.603***	-0.558***
	(0.0986)	(0.127)	(0.0921)
Observations	59,035	59,929	59,121
Panel C: Log Motor Vehicle Theft			
log(MW)	0.112	0.221	0.131
	(0.245)	(0.209)	(0.244)
log(MW) × High $\overline{\text{HHI}}$	-0.180	-0.501**	-0.165
	(0.226)	(0.206)	(0.205)
Observations	51,862	52,361	51,933
Panel D: Log Burglary			
log(MW)	0.302**	0.284**	0.317**
	(0.140)	(0.121)	(0.137)
log(MW) × High $\overline{\text{HHI}}$	-0.232*	-0.217	-0.192
	(0.127)	(0.143)	(0.118)
Observations	55,028	55,626	55,107
County & Time FE	✓	✓	✓
State-specific time trends	✓	✓	✓
Baseline controls	✓	✓	✓
Crime controls	✓	✓	✓

Note: This table presents estimates of the elasticity of employment and property crime with respect to the minimum wage, interacted with a discrete indicator for a highly concentrated local labor market (High HHI, defined as $\overline{\text{HHI}} \geq 0.25$). The estimates are obtained from estimating Equation (5). The dependent variables vary by panel: log retail employment (Panel A), log larceny-theft (Panel B), log motor vehicle theft (Panel C), and log burglary (Panel D). The data are a county-quarter panel spanning 2010–2016. All specifications include county fixed effects, Census-region-by-year-quarter fixed effects, and state-specific linear time trends. Time-varying economic controls include the log of county population, total aggregate employment, total earnings, and the unemployment rate. Criminal justice controls include log violent crime, log clearance rates for property and violent crimes, and the number of reporting police agencies. Standard errors, reported in parentheses, are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.4. Placebo Tests: Minimum Wage Elasticities of Violent Crime by Labor Market Concentration

VARIABLES	(1) Stock Clerks	(2) Retail Sales	(3) Cashiers
Panel A: Continuous HHI			
log(MW)	0.0385	0.0876	-0.0989
	(0.183)	(0.188)	(0.240)
log(MW) $\times$ $\overline{\text{HHI}}$	0.729	1.016	1.066
	(0.536)	(0.803)	(0.684)
Observations	55,291	55,904	55,378
Panel B: Discrete HHI			
log(MW)	0.203	0.297	0.120
	(0.223)	(0.187)	(0.214)
log(MW) $\times$ High $\overline{\text{HHI}}$	0.475	0.523	0.591*
	(0.342)	(0.371)	(0.318)
Observations	61,678	62,440	61,760
County & Time FE	✓	✓	✓
State-specific time trends	✓	✓	✓
Baseline controls	✓	✓	✓
Crime controls (excluding violent crime)	✓	✓	✓

Note: This table presents placebo test estimates of the elasticity of violent crime with respect to the minimum wage, as a function of local labor market concentration. Across all specifications, the dependent variable is log violent crime. Panel A interacts the minimum wage with the continuous measure of local concentration ($\overline{\text{HHI}}$) using Equation (4). Panel B utilizes the discrete indicator (High HHI, defined as $\overline{\text{HHI}} \geq 0.25$) using Equation (5). The data are a county-quarter panel spanning 2010–2016. All specifications include county fixed effects, Census-region-by-year-quarter fixed effects, and state-specific linear time trends. Time-varying economic controls include the log of county population, total aggregate employment, total earnings, and the unemployment rate. Criminal justice controls include log clearance rates for property and violent crimes, and the number of reporting police agencies; log violent crime is excluded as a control here, as it serves as the dependent variable. Standard errors, reported in parentheses, are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.5. Robustness Check: Minimum Wage Elasticities of Employment and Property Crime by Continuous Labor Market Concentration (No Crime Controls)

VARIABLES	(1) Stock Clerks	(2) Retail Sales	(3) Cashiers
Panel A: Log Retail Employment			
log(MW)	-0.271**	-0.221***	-0.288**
	(0.102)	(0.0701)	(0.115)
log(MW) × $\overline{\text{HHI}}$	0.640***	0.837***	0.692***
	(0.203)	(0.217)	(0.249)
Observations	56,953	57,489	56,913
Panel B: Log Larceny-Theft			
log(MW)	0.380**	0.303**	0.477***
	(0.145)	(0.139)	(0.137)
log(MW) × $\overline{\text{HHI}}$	-0.647***	-0.588**	-0.771***
	(0.228)	(0.283)	(0.264)
Observations	63,128	63,888	63,146
Panel C: Log Motor Vehicle Theft			
log(MW)	0.144	0.131	0.182
	(0.275)	(0.210)	(0.253)
log(MW) × $\overline{\text{HHI}}$	-0.320	-0.381	-0.352
	(0.385)	(0.326)	(0.360)
Observations	56,991	57,570	56,999
Panel D: Log Burglary			
log(MW)	0.268	0.250	0.292*
	(0.203)	(0.165)	(0.165)
log(MW) × $\overline{\text{HHI}}$	-0.111	-0.00625	-0.0995
	(0.324)	(0.329)	(0.309)
Observations	62,561	63,299	62,556
County & Time FE	✓	✓	✓
State-specific time trends	✓	✓	✓
Baseline controls	✓	✓	✓
Crime controls			

Note: This table presents estimates of the elasticity of employment and property crime with respect to the minimum wage as a function of local labor market concentration ($\overline{\text{HHI}}$), obtained from estimating Equation (4). The dependent variables vary by panel: log retail employment (Panel A), log larceny-theft (Panel B), log motor vehicle theft (Panel C), and log burglary (Panel D). The data are a county-quarter panel spanning 2010–2016. All specifications include county fixed effects, Census-region-by-year-quarter fixed effects, and state-specific linear time trends. Time-varying economic controls include the log of county population, total aggregate employment, total earnings, and the unemployment rate. This specification excludes criminal justice controls. Standard errors, reported in parentheses, are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.6. Robustness Check: Minimum Wage Elasticities of Employment and Property Crime in Highly Concentrated Labor Markets (No Crime Controls)

VARIABLES	(1) Stock Clerks	(2) Retail Sales	(3) Cashiers
Panel A: Log Retail Employment			
log(MW)	-0.163*	-0.187**	-0.180**
	(0.0868)	(0.0761)	(0.0861)
log(MW) × High $\overline{\text{HHI}}$	0.264***	0.478***	0.299***
	(0.0765)	(0.0924)	(0.0746)
Observations	56,953	57,489	56,913
Panel B: Log Larceny-Theft			
log(MW)	0.406***	0.373***	0.464***
	(0.129)	(0.133)	(0.124)
log(MW) × High $\overline{\text{HHI}}$	-0.373***	-0.389***	-0.399***
	(0.0909)	(0.130)	(0.0848)
Observations	64,006	64,806	64,064
Panel C: Log Motor Vehicle Theft			
log(MW)	0.0883	0.203	0.102
	(0.244)	(0.219)	(0.246)
log(MW) × High $\overline{\text{HHI}}$	-0.124	-0.422**	-0.113
	(0.214)	(0.200)	(0.193)
Observations	56,991	57,570	56,999
Panel D: Log Burglary			
log(MW)	0.278*	0.337**	0.266
	(0.163)	(0.152)	(0.162)
log(MW) × High $\overline{\text{HHI}}$	-0.0891	-0.194	-0.0297
	(0.149)	(0.157)	(0.145)
Observations	62,561	63,299	62,556
County & Time FE	✓	✓	✓
State-specific time trends	✓	✓	✓
Baseline controls	✓	✓	✓
Crime controls			

Note: This table presents estimates of the elasticity of employment and property crime with respect to the minimum wage, interacted with a discrete indicator for a highly concentrated local labor market (High HHI, defined as $\overline{\text{HHI}} \geq 0.25$). The estimates are obtained from estimating Equation (5). The dependent variables vary by panel: log retail employment (Panel A), log larceny-theft (Panel B), log motor vehicle theft (Panel C), and log burglary (Panel D). The data are a county-quarter panel spanning 2010–2016. All specifications include county fixed effects, Census-region-by-year-quarter fixed effects, and state-specific linear time trends. Time-varying economic controls include the log of county population, total aggregate employment, total earnings, and the unemployment rate. This specification excludes criminal justice controls. Standard errors, reported in parentheses, are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.7. Robustness Check: Minimum Wage Elasticities of Employment and Property Crime by Continuous Labor Market Concentration (Additional Demographic Controls)

VARIABLES	(1) Stock Clerks	(2) Retail Sales	(3) Cashiers
Panel A: Log Retail Employment			
log(MW)	-0.288***	-0.247***	-0.299**
	(0.106)	(0.0812)	(0.116)
log(MW) × $\overline{\text{HHI}}$	0.676***	0.991***	0.700***
	(0.231)	(0.237)	(0.249)
Observations	44,286	44,459	44,334
Panel B: Log Larceny-Theft			
log(MW)	0.370**	0.236*	0.464***
	(0.151)	(0.126)	(0.140)
log(MW) × $\overline{\text{HHI}}$	-0.864***	-0.708***	-0.975***
	(0.193)	(0.206)	(0.210)
Observations	55,218	55,829	55,310
Panel C: Log Motor Vehicle Theft			
log(MW)	0.115	0.0976	0.127
	(0.259)	(0.203)	(0.246)
log(MW) × $\overline{\text{HHI}}$	-0.407	-0.499	-0.369
	(0.364)	(0.364)	(0.360)
Observations	51,831	52,330	51,902
Panel D: Log Burglary			
log(MW)	0.303*	0.211*	0.384***
	(0.159)	(0.123)	(0.133)
log(MW) × $\overline{\text{HHI}}$	-0.434	-0.197	-0.521**
	(0.280)	(0.262)	(0.197)
Observations	54,994	55,592	55,073
County & Time FE	✓	✓	✓
State-specific time trends	✓	✓	✓
Baseline controls	✓	✓	✓
Crime controls	✓	✓	✓
Demographic controls	✓	✓	✓

Note: This table presents estimates of the elasticity of employment and property crime with respect to the minimum wage as a function of local labor market concentration ($\overline{\text{HHI}}$), obtained from estimating Equation (4). The dependent variables vary by panel: log retail employment (Panel A), log larceny-theft (Panel B), log motor vehicle theft (Panel C), and log burglary (Panel D). The data are a county-quarter panel spanning 2010–2016. All specifications include county fixed effects, Census-region-by-year-quarter fixed effects, and state-specific linear time trends. Time-varying economic controls include the log of county population, total aggregate employment, total earnings, and the unemployment rate. Criminal justice controls include log violent crime, log clearance rates for property and violent crimes, and the number of reporting police agencies. This specification additionally includes demographic controls for county-level age, sex, and race distributions. Standard errors, reported in parentheses, are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.8. Robustness Check: Minimum Wage Elasticities of Employment and Property Crime in Highly Concentrated Labor Markets (Additional Demographic Controls)

VARIABLES	(1) Stock Clerks	(2) Retail Sales	(3) Cashiers
Panel A: Log Retail Employment			
log(MW)	-0.161*	-0.191**	-0.186**
	(0.0818)	(0.0760)	(0.0808)
log(MW) × High $\overline{\text{HHI}}$	0.213**	0.499***	0.266***
	(0.0816)	(0.107)	(0.0804)
Observations	44,286	44,459	44,334
Panel B: Log Larceny-Theft			
log(MW)	0.326**	0.231*	0.385***
	(0.138)	(0.121)	(0.136)
log(MW) × High $\overline{\text{HHI}}$	-0.508***	-0.457***	-0.529***
	(0.0936)	(0.0967)	(0.0926)
Observations	55,218	55,829	55,310
Panel C: Log Motor Vehicle Theft			
log(MW)	0.0554	0.162	0.0554
	(0.239)	(0.205)	(0.232)
log(MW) × High $\overline{\text{HHI}}$	-0.172	-0.495**	-0.134
	(0.218)	(0.195)	(0.186)
Observations	51,831	52,330	51,902
Panel D: Log Burglary			
log(MW)	0.302**	0.260**	0.305**
	(0.130)	(0.117)	(0.129)
log(MW) × High $\overline{\text{HHI}}$	-0.288**	-0.248**	-0.228**
	(0.109)	(0.122)	(0.0980)
Observations	54,994	55,592	55,073
County & Time FE	✓	✓	✓
State-specific time trends	✓	✓	✓
Baseline controls	✓	✓	✓
Crime controls	✓	✓	✓
Demographic controls	✓	✓	✓

Note: This table presents estimates of the elasticity of employment and property crime with respect to the minimum wage, interacted with a discrete indicator for a highly concentrated local labor market (High HHI, defined as $\overline{\text{HHI}} \geq 0.25$). The estimates are obtained from estimating Equation (5). The dependent variables vary by panel: log retail employment (Panel A), log larceny-theft (Panel B), log motor vehicle theft (Panel C), and log burglary (Panel D). The data are a county-quarter panel spanning 2010–2016. All specifications include county fixed effects, Census-region-by-year-quarter fixed effects, and state-specific linear time trends. Time-varying economic controls include the log of county population, total aggregate employment, total earnings, and the unemployment rate. Criminal justice controls include log violent crime, log clearance rates for property and violent crimes, and the number of reporting police agencies. This specification additionally includes demographic controls for county-level age, sex, and race distributions. Standard errors, reported in parentheses, are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.9. Robustness Check: Minimum Wage Elasticities of Employment and Property Crime by Continuous Labor Market Concentration (Pre-determined HHI)

VARIABLES	(1) Stock Clerks	(2) Retail Sales	(3) Cashiers
Panel A: Log Retail Employment			
log(MW)	-0.271**	-0.231**	-0.279**
	(0.105)	(0.0915)	(0.124)
log(MW) × $\overline{\text{HHI}}$	0.386***	0.425***	0.405*
	(0.142)	(0.140)	(0.210)
Observations	37,940	42,022	38,250
Panel B: Log Larceny-Theft			
log(MW)	0.686***	0.394**	0.721***
	(0.170)	(0.160)	(0.158)
log(MW) × $\overline{\text{HHI}}$	-0.981***	-0.810***	-1.072***
	(0.228)	(0.184)	(0.161)
Observations	43,454	50,351	44,175
Panel C: Log Motor Vehicle Theft			
log(MW)	0.00571	0.153	0.0869
	(0.349)	(0.246)	(0.329)
log(MW) × $\overline{\text{HHI}}$	-0.126	-0.376	-0.353
	(0.444)	(0.305)	(0.421)
Observations	41,786	48,025	42,509
Panel D: Log Burglary			
log(MW)	0.172	0.302**	0.349*
	(0.177)	(0.142)	(0.186)
log(MW) × $\overline{\text{HHI}}$	-0.0423	-0.474*	-0.334
	(0.251)	(0.260)	(0.262)
Observations	43,354	50,219	44,086
County & Time FE	✓	✓	✓
State-specific time trends	✓	✓	✓
Baseline controls	✓	✓	✓
Crime controls	✓	✓	✓

Note: This table presents estimates of the elasticity of employment and property crime with respect to the minimum wage as a function of a predetermined measure of local labor market concentration ($\overline{\text{HHI}}$), obtained from estimating Equation (4). To address potential endogeneity, $\overline{\text{HHI}}$ is calculated using online vacancy data from 2010–2011, and the regression is estimated on a county-quarter panel spanning 2012–2016. The dependent variables vary by panel: log retail employment (Panel A), log larceny-theft (Panel B), log motor vehicle theft (Panel C), and log burglary (Panel D). Columns (1) through (3) define local concentration using the predetermined HHI for Stock Clerks, Retail Salespersons, and Cashiers, respectively. All specifications include county fixed effects, Census-region-by-year-quarter fixed effects, and state-specific linear time trends. Time-varying economic controls include the log of county population, total aggregate employment, total earnings, and the unemployment rate. Criminal justice controls include log violent crime, log clearance rates for property and violent crimes, and the number of reporting police agencies. Standard errors, reported in parentheses, are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.10. Robustness Check: Minimum Wage Elasticities of Employment and Property Crime in Highly Concentrated Labor Markets (Pre-determined HHI)

VARIABLES	(1) Stock Clerks	(2) Retail Sales	(3) Cashiers
Panel A: Log Retail Employment			
log(MW)	-0.155*	-0.162**	-0.146
	(0.0899)	(0.0772)	(0.0933)
log(MW) × High $\overline{\text{HHI}}$	0.145**	0.205**	0.130*
	(0.0554)	(0.0788)	(0.0684)
Observations	37,940	42,022	38,250
Panel B: Log Larceny-Theft			
log(MW)	0.357**	0.372**	0.436**
	(0.152)	(0.160)	(0.168)
log(MW) × High $\overline{\text{HHI}}$	-0.339***	-0.599***	-0.446***
	(0.105)	(0.120)	(0.110)
Observations	43,454	50,351	44,175
Panel C: Log Motor Vehicle Theft			
log(MW)	-0.368	0.0884	-0.272
	(0.348)	(0.243)	(0.376)
log(MW) × High $\overline{\text{HHI}}$	0.363	-0.182	0.170
	(0.288)	(0.239)	(0.338)
Observations	41,786	48,025	42,509
Panel D: Log Burglary			
log(MW)	0.202	0.240*	0.380**
	(0.178)	(0.136)	(0.175)
log(MW) × High $\overline{\text{HHI}}$	-0.0675	-0.270*	-0.278
	(0.156)	(0.148)	(0.173)
Observations	43,354	50,219	44,086
County & Time FE	✓	✓	✓
State-specific time trends	✓	✓	✓
Baseline controls	✓	✓	✓
Crime controls	✓	✓	✓

Note: This table presents estimates of the elasticity of employment and property crime with respect to the minimum wage, interacted with a discrete indicator for a highly concentrated local labor market (High HHI, defined as $\overline{\text{HHI}} \geq 0.25$). The estimates are obtained from estimating Equation (5). To address potential endogeneity, the concentration indicator is predetermined using online vacancy data from 2010–2011, and the regression is estimated on a county-quarter panel spanning 2012–2016. The dependent variables vary by panel: log retail employment (Panel A), log larceny-theft (Panel B), log motor vehicle theft (Panel C), and log burglary (Panel D). Columns (1) through (3) define local concentration using the predetermined discrete HHI for Stock Clerks, Retail Salespersons, and Cashiers, respectively. All specifications include county fixed effects, Census-region-by-year-quarter fixed effects, and state-specific linear time trends. Time-varying economic controls include the log of county population, total aggregate employment, total earnings, and the unemployment rate. Criminal justice controls include log violent crime, log clearance rates for property and violent crimes, and the number of reporting police agencies. Standard errors, reported in parentheses, are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.